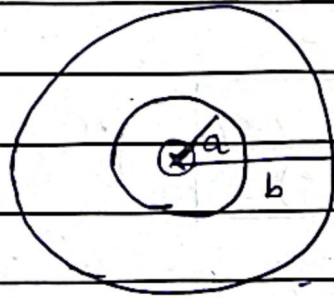


Homework 5

Syed M. Asad Ali (15a09475)

Ampere's Circuit Law:

7.18 b) A hollow conducting cylinder has inner radius a and outer radius b and carries current I along the positive z -direction. Find $\vec{H}$ everywhere.



1) for $\rho < a$; $I_{enc} = 0 \rightarrow \boxed{\vec{H} = 0}$

2) for $\rho > b$; $I_{enc} = I = H 2\pi \rho \rightarrow \boxed{\vec{H} = \frac{I}{2\pi \rho} \hat{\phi}}$

3) for $a < \rho < b$;

$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

$$\therefore J = \frac{\Delta I}{\Delta S} = \frac{I}{\pi(b^2 - a^2)}$$

$$I_{en} = \int_{\phi=0}^{2\pi} \int_{\rho=a}^{\rho} \frac{I}{\pi(b^2 - a^2)} \cdot \rho d\phi d\rho$$

$$I_{enc} = \frac{I}{\pi(b^2 - a^2)} \cdot \frac{\rho^2}{2} \Big|_a^{\rho} \cdot \phi \Big|_0^{2\pi}$$

$$I_{enc} = I \frac{1}{2\pi(b^2 - a^2)} \cdot 2\pi$$

$$I_{enc} = \frac{I(p^2 - a^2)}{b^2 - a^2}$$

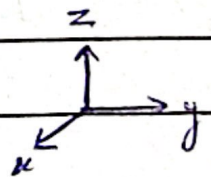
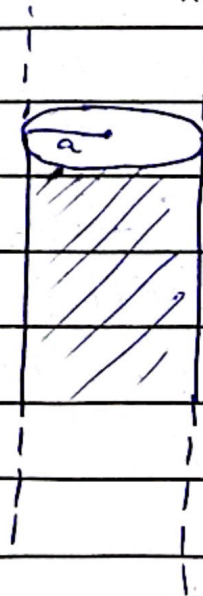
$$LHS = RHS$$

$$2\pi p H = I \frac{(p^2 - a^2)}{b^2 - a^2}$$

$$\boxed{H = \frac{I(p^2 - a^2) \cdot a\phi}{2\pi p(b^2 - a^2)}} \quad \text{for } a < p < b.$$

7-21) a) An infinitely long solid conductor of radius a is placed along the z -axis. If conductor carries current I in $+z$ direction. Show

$$H = \frac{I p}{2\pi a^2} \cdot a\phi$$



LHS: {outside conductor}

$$I_{enc} = \oint \vec{H} \cdot d\vec{l}$$

$$I_{enc} = \int_0^{2\pi} H \cdot \rho d\phi \Rightarrow H \rho \int_0^{2\pi} d\phi \Rightarrow \boxed{2\pi \rho H} \quad \rho \geq a$$

RHS: {Inside conductor}

$$\vec{J} = \frac{I}{\pi a^2} \cdot a\hat{z}, \quad d\vec{S}' = \rho' d\phi' d\rho' \cdot a\hat{z}$$

$$I_{enc} = \int_S \vec{J} \cdot d\vec{S}$$

$$= \int_0^{2\pi} \int_0^{\rho} \frac{I}{\pi a^2} \cdot \rho' d\phi' d\rho'$$

$$= \frac{I}{\pi a^2} \cdot \left[\frac{\rho'^2}{2} \right]_0^{\rho} \cdot \left[\phi' \right]_0^{2\pi}$$

$$\boxed{I_{enc} = \frac{I \rho^2}{a^2}}$$

since; LHS = RHS

$$2\pi \rho H = \frac{I \rho^2}{a^2}$$

$$H = \frac{I \rho}{2\pi a^2}$$

$$\frac{I \rho}{2\pi a^2}$$

$$\boxed{H = \frac{I \rho}{2\pi a^2} \cdot a\hat{\phi}} \quad \rho \leq a$$

b) If $I = 3A$ and $a = 2cm$ find $\vec{H}$ at $(0, 1cm, 0)$ and $(0, 4cm, 0)$.

at $(0, 1cm, 0)$:

$\rho = 1cm$

$\rightarrow \rho < a$

$\vec{H} = \frac{I}{2\pi\rho} a^2 = \frac{3}{2\pi(1 \times 10^{-2})} = \frac{300}{2\pi} A/m$

at $(0, 1cm, 0)$:

$\rho = 1cm$

$\rightarrow \rho < a$

$\vec{H} = \frac{I\rho}{2\pi a^2} = \frac{3(1 \times 10^{-2})}{2\pi(2 \times 10^{-2})^2} = \frac{300}{8\pi} A/m$

at $(0, 4cm, 0)$:

$\rho = 4cm$

$\rightarrow \rho > a$

$H = \frac{I}{2\pi\rho} = \frac{3}{2\pi(4 \times 10^{-2})} = \frac{300}{8\pi} A/m$

7-22) An infinitely long cylinder of radius a is placed along z -axis. If current density is $\vec{J} = J_0 \frac{a}{\rho} \hat{z}$, where J_0 is constant, find $\vec{H}$ everywhere.

for $\rho > a$:

$$\oint \vec{H} \cdot d\vec{l} = I_{enc} = \int_V \vec{J} \cdot d\vec{V}$$

$$H 2\pi \rho = \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\rho} \frac{J_0}{\rho} \cdot \rho d\phi d\rho$$

$$H 2\pi \rho = 2\pi \rho J_0$$

$$H = J_0$$

$$\boxed{H = J_0}$$

for $\rho < a$:

$$2\pi \rho H = \int_{\phi=0}^{2\pi} \int_{\rho=0}^{\rho} \frac{J_0}{\rho} \cdot \rho d\phi d\rho$$

$$2\pi \rho H = 2\pi \rho J_0$$

$$\boxed{H = J_0 \frac{a}{\rho}}$$

Magnetic Vector Potential:

7.46) An infinitely long conductor of radius a with $\vec{J} = J_0 \hat{a}_z$. Show that magnetic vector potential for $\rho < a$ is:

$$\vec{A} = -\frac{1}{4} \mu_0 J_0 \rho^2 \hat{a}_z$$

1) Find $\vec{H}$

2) Find $\vec{B}$ $\{ \vec{B} = \mu_0 \vec{H} \}$

3) Find $\vec{A}$ $\{ \vec{B} = \nabla \times \vec{A} \}$

$$2\pi \rho H = \int_0^{2\pi} \int_0^{\rho} J_0 \cdot \rho' d\phi d\rho'$$

$$2\pi \rho H = J_0 \cdot \frac{\rho^2}{2} \Big|_0^{\rho} \cdot \phi \Big|_0^{2\pi}$$

$$2\pi \rho H = 2\pi J_0 \cdot \frac{\rho^2}{2}$$

$$\rightarrow H = \frac{J_0 \cdot \rho}{2} \cdot \hat{a}_\phi$$

$$\rightarrow \vec{B} = \mu_0 \vec{H} = \frac{J_0 \mu_0 \rho}{2} \cdot \hat{a}_\phi$$

$$\rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\frac{J_0 \mu_0 \rho}{2} \hat{a}_\phi = -\frac{\partial A_z}{\partial \rho} \hat{a}_\phi$$

$$\frac{\partial A_z}{\partial \rho} = -\frac{J_0 \mu_0 \rho}{2}$$

$$\nabla \times \vec{A} = \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{a}_\rho +$$

$$\left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{a}_\phi +$$

$$\frac{1}{\rho} \left(\frac{\partial A_\phi}{\partial \rho} - \frac{\partial A_\rho}{\partial \phi} \right) \hat{a}_z$$

Date: _____

$$\bar{A} = -\frac{J_0 \mu_0}{2} \frac{\rho'^2}{2} \Big|_0^\rho$$

$$\bar{A} = -\frac{1}{4} \mu_0 J_0 \rho^2 \cdot a_z$$

7.47) In free space, $\bar{A} = 10 \sin \pi x \cdot y a_x + (4 + \cos \pi x) a_z$ Wb/m.
Find $\bar{H}$ and $\bar{J}$.

$$\bar{B} = \nabla \times \bar{A} = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 10 \sin \pi x \cdot y & 0 & 4 + \cos \pi x \end{vmatrix}$$

$$= \frac{\partial}{\partial y} (4 + \cos \pi x) \cdot a_x - \frac{\partial}{\partial x} (4 + \cos \pi x) + \frac{\partial}{\partial z} (10 \sin \pi x y) a_y$$

$$= -\frac{\partial}{\partial y} (10 \sin \pi x)$$

$$\rightarrow \bar{B} = \pi \sin \pi x \cdot a_y - 10 \pi \cos \pi x \cdot a_z$$

$$\rightarrow \bar{H} = \frac{\bar{B}}{\mu_0} = \frac{\pi}{\mu_0} (\sin \pi x \cdot a_y - 10 \cos \pi x \cdot a_z)$$

$$\rightarrow \bar{J} = \nabla \times \bar{H}$$

	a_x	a_y	a_z
$\vec{J} = \frac{\pi}{\mu_0}$	$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial}{\partial z}$
	0	$\sin \pi x$	$-10 \cos \pi y$

$$\vec{J} = \frac{\pi}{\mu_0} \left[\frac{\partial (-10 \cos \pi y)}{\partial y} - \frac{\partial (\sin \pi x)}{\partial z} \right] a_x - \left[\frac{\partial (-10 \cos \pi y)}{\partial x} \right] a_y + \left[\frac{\partial (\sin \pi x)}{\partial x} \right] a_z$$

$$\vec{J} = \frac{\pi}{\mu_0} \left[10 \pi \sin \pi y \cdot a_x + \pi \cos \pi x \cdot a_z \right]$$

$$\vec{J} = \frac{\pi^2}{\mu_0} \left[10 \sin \pi y \cdot a_x + \cos \pi x \cdot a_z \right]$$

7.48) The magnetic vector potential at a distant point from a small circular loop is given by;

$$\vec{A} = \frac{A_0}{r^2} \sin \theta \cdot a_\phi \quad \text{wb/m}$$

Find $\vec{B} = ?$

* i took ρ for r accidently.

$$\vec{B} = \nabla \times \vec{A}$$

$$\therefore \nabla \times \vec{A} = \left(\frac{1}{\rho \sin \theta} \frac{\partial (\sin \theta \cdot A_\phi)}{\partial \theta} - \frac{1}{\rho \sin \theta} \frac{\partial A_\theta}{\partial \phi} \right) a_\rho +$$

$$\left(\frac{1}{r} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial A_\rho}{\partial \theta} \right) - \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) \right) a_\theta +$$

$$\left(\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\theta) - \frac{1}{r} \frac{\partial A_\rho}{\partial \theta} \right) a_\phi.$$

$$\vec{B} = \frac{1}{\rho \sin \theta} \frac{\partial (\sin \theta \cdot A_\phi)}{\partial \theta} a_\rho - \frac{1}{\rho} \frac{\partial (\rho A_\phi)}{\partial \rho} a_\theta.$$

$$\vec{B} = \frac{1}{\rho \sin \theta} \frac{\partial (\sin \theta \cdot \sin \theta A_0)}{\partial \theta} a_\rho - \frac{1}{\rho} \frac{\partial (\rho \cdot A_0 \sin \theta)}{\partial \rho} a_\theta.$$

$$= * \boxed{\rho = r}$$

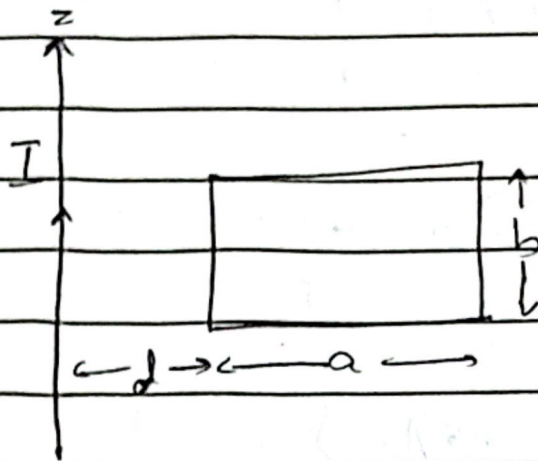
$$\vec{B} = \frac{1}{r \sin \theta} \frac{\partial (\sin^2 \theta)}{\partial \theta} A_0 a_\rho - \cancel{\frac{1}{r} \frac{\partial (\rho A_0 \sin \theta)}{\partial \rho} a_\theta} \left(\frac{1}{r} \frac{\partial (\rho A_0 \sin \theta)}{\partial \rho} a_\theta \right)$$

$$= \frac{A_0 (2 \sin \theta \cos \theta)}{r^3 \sin \theta} a_\rho + \frac{A_0 \sin \theta}{r^3} a_\theta.$$

$$\boxed{\vec{B} = \frac{A_0}{r^3} (2 \cos \theta \cdot a_\rho + \sin \theta \cdot a_\theta)}$$

Application - level problems:

7.29) Determine magnetic flux through rectangular loop ($a \times b$) due to infinitely long conductor carrying current I . The loop and the straight conductors are separated by distance d .



$$\vec{H} = \frac{I}{2\pi\rho} \cdot a\phi$$

$$\vec{B} = \frac{I\mu_0}{2\pi\rho} \cdot a\phi$$

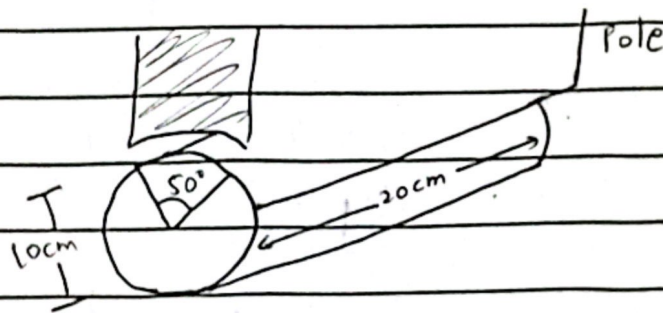
$$\begin{aligned} \text{Magnetic flux } (\Psi) &= \int \vec{B} \cdot d\vec{S} \\ &= \int_{\rho=d}^{a+d} \int_{z=0}^b \frac{\mu_0 I}{2\pi\rho} \cdot d\rho dz \\ &= \frac{I\mu_0}{2\pi} \ln(\rho) \Big|_d^{a+d} \cdot z \Big|_0^b \end{aligned}$$

$$\psi = \frac{\mu_0 I \cdot b}{2\pi} \ln \left[\frac{d+a}{d} \right]$$

7-33) The electric motor shown in Fig. 7.40. has field.

$$\vec{H} = \frac{10^6}{\rho} \sin 2\phi \text{ a}_\rho \text{ A/m.}$$

calculate flux per pole passing through the air gap if axial length of pole is 20 cm



$$\vec{B} = \mu_0 \cdot 10^6 \sin 2\phi \text{ a}_\rho$$

$$\psi = \int_{z=0}^{0.2} \int_{\phi=0}^{50^\circ} \frac{\mu_0 \cdot 10^6 \sin 2\phi}{\rho} \cdot \rho d\phi dz$$

$$\psi = \mu_0 \cdot 10^6 \cdot \left. \frac{-\cos 2\phi}{2} \right|_0^{50^\circ} \cdot z \Big|_0^{0.2}$$

$$\psi = \mu_0 \cdot 10^6 (1 - \cos 100^\circ) (0.2)$$

$$\psi = (4\pi \times 10^{-7} \cdot 10^6) (0.2) (1 - \cos 100^\circ)$$

$$\psi = 0.1475 \text{ wb}$$